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SEMICONDUCTOR DEVICE

Abstract

A semiconductor device may include an active region extending in a first direction, a gate structure on the active region and extending in a second direction, a source/drain region on the active region and a side of the gate structure, a contact structure on and electrically connected to the source/drain region, a via structure on and electrically connected to the contact structure, a gate contact structure on and electrically connected to the gate structure, a first signal transmission line structure on the gate structure, and a power transmission line structure on the contact structure. The first signal transmission line structure may include a first line portion extending across the gate structure and a first protrusion portion vertically protruding toward the gate contact structure. The power transmission line structure may include an interconnection portion extending across the gate and contact structures, and an extension portion vertically extending toward the via structure.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION(S)

[0001] This application claims benefit of priority to Korean Patent Application No. 10-2024-0026664 filed on Feb. 23, 2024 in the Korean Intellectual Property Office, the disclosure of which is incorporated herein by reference in its entirety.

BACKGROUND

[0002] Inventive concepts relate to a semiconductor device.

[0003] As demand for implementation of high performance, high speed, and/or multi-functionalization of semiconductor devices increases, a degree of integration of semiconductor devices has been increasing. In manufacturing semiconductor devices having a fine pattern corresponding to the trend for a high degree of integration of semiconductor devices, it may be necessary to implement patterns having a fine width or a fine separation distance.

SUMMARY

[0004] An aspect of inventive concepts provides a semiconductor device including interconnection lines respectively having a step in one direction to limit and/or prevent a short circuit between the interconnection lines.

[0005] According to an example embodiment of inventive concepts, a semiconductor device may include a substrate including an active region extending in a first direction; a gate structure on the active region and intersecting the active region, the gate structure including a gate electrode and a gate capping layer on the gate electrode, the gate structure extending in a second direction, the second direction intersecting the first direction; a source/drain region on the active region, the source/drain region on a side of the gate structure; a contact structure on the source/drain region and electrically connected to the source/drain region; a first insulating layer on the gate structure and the contact structure; a gate contact structure passing through the first insulating layer and the gate capping layer of the gate structure, the gate contact structure electrically connected to the gate electrode of the gate structure; a via structure passing through the first insulating layer, the via structure being electrically connected to the contact structure; an etch stop layer on the first insulating layer and the gate contact structure; a second insulating layer on the etch stop layer; an insulating liner on the second insulating layer; and interconnection lines on the first insulating layer, the interconnection lines including a first line, a second line, and a third line sequentially spaced apart in the second direction on the first insulating layer. Each of the first line, the second line, and the third line may include a line portion and an extension portion extending from the line portion in a third direction. In each of the first line, the second line, and the third line, the line portion may pass through the insulating liner and extend in the first direction on the second insulating layer. The third direction may be perpendicular to the first direction and the second direction. The extension portion of the first line, the extension portion of the second line, and the extension portion of the third line each may pass through the second insulating layer and the etch stop layer and may be electrically connected to a corresponding one of the gate contact structure and the via structure.

[0006] According to an example embodiment of inventive concepts, a semiconductor device may

include a substrate including an active region extending in a first direction; a gate structure on the active region and intersecting the active region, the gate structure extending in a second direction, the second direction intersecting the first direction; a source/drain region on the active region, the source/drain region on a side of the gate structure; a contact structure on the source/drain region and electrically connected to the source/drain region; a via structure on the contact structure and electrically connected to the contact structure; a gate contact structure on the gate structure and electrically connected to the gate structure; a first signal transmission line structure including a first line portion extending in one direction across the gate structure and a first protrusion portion vertically protruding from a lower surface of the first line portion toward the gate contact structure, the first signal transmission line structure being on the gate structure; and a power transmission line structure including an interconnection portion and an extension portion, the interconnection portion extending in the one direction across the gate structure and the contact structure, and the extension portion vertically extending from a lower surface of the interconnection portion toward the via structure, the power transmission line structure being on the contact structure.

[0007] According to an example embodiment of inventive concepts, a semiconductor device may include a substrate including an active region extending in a first direction; a gate structure on the active region and intersecting the active region, the gate structure including a gate electrode and a gate capping layer on the gate electrode, the gate structure extending in a second direction, the second direction intersecting the first direction; a source/drain region on the active region, the source/drain region on a side of the gate structure; a contact structure on the source/drain region and electrically connected to the source/drain region; a first insulating layer on the gate structure and the contact structure; a gate contact structure passing through the first insulating layer and the gate capping layer of the gate structure, the gate contact structure electrically connected to the gate electrode of the gate structure; a via structure passing through the first insulating layer, the via structure electrically connected to the contact structure; an etch stop layer on the first insulating layer and the gate contact structure; a second insulating layer on the etch stop layer; an insulating liner on the second insulating layer; and first interconnection lines on the first insulating layer and spaced apart from each other in the second direction. The first interconnection lines may include a first line structure and a second line structure. The first line structure may include a first line portion and a protrusion portion. The first line portion may be on the second insulating layer and extend in one direction across the gate structure. The protrusion portion may vertically protrude from a lower surface of the first line portion, pass through the second insulating layer and the etch stop layer, and may be in contact with the gate contact structure. The second line structure may include an interconnection portion and an extension portion. The interconnection portion may be on the second insulating layer and may extend in the one direction across the gate structure and the contact structure. The extension portion may vertically protrude from a lower surface of the interconnection portion, pass through the second insulating layer and the etch stop layer, and may be in contact with the via structure. Outer peripheries of the protrusion portion of the first line structure and the extension portion of the second line structure may have a circular cross-section.

Description

BRIEF DESCRIPTION OF DRAWINGS

[0008] The above and other aspects, features, and advantages of inventive concepts will be more clearly understood from the following detailed description, taken in conjunction with the accompanying drawings, in which:

[0009] FIG. 1 is a plan view of a semiconductor device according to an example embodiment;

[0010] FIG. 2 is a cross-sectional view of a semiconductor device according to example embodiments;

[0011] FIG. 3 is a partially enlarged view of a semiconductor device according to example embodiments;

[0012] FIG. 4 is a cross-sectional view of a semiconductor device according to example embodiments;

[0013] FIG. 5 is a partially enlarged view of a semiconductor device according to example embodiments;

[0014] FIG. 6 is a cross-sectional view of a semiconductor device according to example embodiments;

[0015] FIGS. 7 to 11 are partially enlarged views of a semiconductor device according to example embodiments;

[0016] FIG. 12 is a cross-sectional view of a semiconductor device according to example embodiments; and

[0017] FIGS. 13 to 22 are vertical cross-sectional views of sequential processes of a method of manufacturing a semiconductor device according to an example embodiment.

DETAILED DESCRIPTION

[0018] Hereinafter, example embodiments of inventive concepts are described with reference to specific example embodiments and accompanying drawings.

[0019] Inventive concepts may, however, be embodied in many different forms and should not be construed as being limited to the specific example embodiments set forth herein. In addition, example embodiments of inventive concepts may be provided for a more complete description of inventive concepts to those skilled in the art. Accordingly, the shapes and sizes of the elements in the drawings may be exaggerated for clarity of description, and elements denoted by the same reference numerals in the drawings may be the same elements.

[0020] The terminology used herein is for the purpose of describing particular example embodiments only and is not to be limiting of the example embodiments. As used herein, the singular forms “a,” “an,” and “the” are intended to include the plural forms as well, unless the context clearly indicates otherwise. It will be further understood that the terms “comprises” and/or “comprising,” when used in this code, specify the presence of stated features, integers, steps, operations, elements, components or a combination thereof, but do not preclude the presence or addition of one or more other features, integers, steps, operations, elements, components, and/or groups thereof. As used herein, the term “and/or” includes any one and any combination of any two or more of the associated listed items.

[0021] While the term “equal to” is used in the description of example embodiments, it should be understood that some imprecisions may exist. Thus, when one element is referred to as “equal to” another element, it should be understood that an element or a value may be “equal to” another element within a desired manufacturing or operational tolerance range (e.g., $\pm 10\%$).

[0022] The notion that elements are “substantially the same” may indicate that the element may be completely the same and may also indicate that the elements may be determined to be the same in consideration of errors or deviations occurring during a process.

[0023] It will be understood that, although the terms “first,” “second,” “third,” and the like may be used herein to describe various elements, components, regions, layers and/or sections, these elements, components, regions, layers and/or sections should not be limited by these terms. These terms are only used to distinguish one element, component, region, layer or section from another element, component, region, layer or section. Thus, a first element, component, region, layer or section discussed below could be termed a second element, component, region, layer or section without departing from the teachings of inventive concepts.

[0024] When the terms “about” or “substantially” are used in this specification in connection with a numerical value, it is intended that the associated numerical value includes a manufacturing or operational tolerance (e.g., $\pm 10\%$) around the stated numerical value. Moreover, when the words “generally” and “substantially” are used in connection with geometric shapes, it is intended that

precision of the geometric shape is not required but that latitude for the shape is within the scope of the disclosure. Further, regardless of whether numerical values or shapes are modified as “about” or “substantially,” it will be understood that these values and shapes should be construed as including a manufacturing or operational tolerance (e.g., $\pm 10\%$) around the stated numerical values or shapes. When ranges are specified, the range includes all values therebetween such as increments of 0.1%.

[0025] FIG. 1 is a plan view of a semiconductor device according to an example embodiment. FIG. 1 may be a plan view of a region corresponding to a standard cell, among cells included in a semiconductor device.

[0026] FIG. 2 is a cross-sectional view of a semiconductor device according to example embodiments. FIG. 2 illustrates cross-sections of the semiconductor device of FIG. 1, taken along lines I-I' and II-II'.

[0027] FIG. 3 is a partially enlarged view of a semiconductor device according to example embodiments. FIG. 3 illustrates partially enlarged views of region “A” and region “B” of FIG. 2.

[0028] FIG. 4 is a cross-sectional view of a semiconductor device according to example embodiments. FIG. 4 illustrates cross-sections of the semiconductor device of FIG. 1, taken along lines III-III' and IV-IV'.

[0029] FIG. 5 is a partially enlarged view of a semiconductor device according to example embodiments. FIG. 5 illustrates partially enlarged views of region “E” and region “F” of FIG. 4.

[0030] FIG. 6 is a cross-sectional view of a semiconductor device according to example embodiments. FIG. 6 illustrates cross-sections of the semiconductor device of FIG. 1, taken along line V-V'.

[0031] For ease of description, FIGS. 1 to 6 illustrate only some components of the semiconductor device.

[0032] Referring to FIGS. 1 to 6, a semiconductor device **100** may include a substrate **101**, an active region ACT extending in a first direction (for example, an X-direction), gate lines GL extending in a second direction (for example, a Y-direction), contact structures CNT connected to the active regions ACT and the gate lines GL, lower vias **V0** connected to the contact structures CNT, and first interconnection lines **M1** connected to the lower vias **V0**. The semiconductor device **100** may further include device isolation layers **110**, a lower interlayer insulating layer **130**, a plurality of upper interlayer insulating layers **150**, at least one etch stop layer **160**, and a liner **170**. In some example embodiments, the semiconductor device **100** may further include second interconnection lines disposed on upper portions of the first interconnection lines **M1**, the second interconnection lines electrically connecting the first interconnection lines **M1** to each other.

[0033] The substrate **101** may have an upper surface extending in the X-direction and the Y-direction. The substrate **101** may include a semiconductor material, such as a group IV semiconductor, a group III-V compound semiconductor, or a group II-VI compound semiconductor. For example, the group IV semiconductor may include silicon, germanium, or silicon-germanium. The substrate **101** may be provided as a bulk wafer, an epitaxial layer, a silicon on insulator (SOI) layer, or a semiconductor on insulator (SeOI) layer.

[0034] The device isolation layers **110** may define active fins in the substrate **101**. The device isolation layers **110** may be formed of an insulating material. The device isolation layers **110** may be formed using, for example, a shallow trench isolation (STI) process. The device isolation layers **110** may be, for example, oxide, nitride, or a combination thereof.

[0035] The active regions ACT may be defined by, for example, the device isolation layers **110** within the substrate **101**, and may include one or more active fins extending in the first direction, for example, the X-direction. The active fins may have a structure of an active fin protruding from the substrate **101**. The active fins may be formed as part of the substrate **101**, or may include an epitaxial layer grown from the substrate **101**. However, on a side surface of the gate structure **140**, the active fins on the substrate **101** may be recessed and source/drain regions **120** may be disposed.

[0036] The source/drain regions **120** may be disposed on the active region ACT, on opposite sides of the gate structure **140**. The source/drain regions **120** may serve as a source region or a drain region of the semiconductor device **100**. The source/drain regions **120** may be in the form of an elevated source/drain having an upper surface, positioned to be higher than a lower surface of the gate structure **140**, but inventive concepts are not limited thereto. For example, the upper surface of each of the source/drain regions **120** may be positioned to be lower than the lower surface of the gate structure **140** (not illustrated). In the present example embodiment, the source/drain regions **120** are illustrated as having a pentagonal shape, but the source/drain regions **120** may have various shapes, for example, one shape among a polygonal shape, a circular shape, and a rectangular shape. In addition, in the present example embodiment, the source/drain regions **120** are illustrated as being disposed on a single active fin, but inventive concepts are not limited thereto. For example, the source/drain regions **120** may be connected or merged to each other on multiple active fins (not illustrated). The source/drain regions **120** may include, for example, silicon or silicon germanium (SiGe).

[0037] The gate lines GL extend in the second direction, for example, the Y-direction, and may be disposed to be spaced apart from each other in the first direction, for example, the X-direction (FIG. **1**). The gate lines GL may include gate electrodes GL_G and dummy gate electrodes GL_D, providing a semiconductor device. For example, the gate lines GL_G, disposed on edges (or boundaries) of a standard cell region, may be the dummy gate electrodes GL_D.

[0038] The gate structures **140** may be disposed on an upper portion of the active region ACT to intersect the active fins and to extend in the second direction, for example, the Y-direction (FIG. **2**). The gate structures **140** may be disposed to correspond to the gate electrodes GL_G of FIG. **1**. The gate structure **140** may include a gate dielectric layer **142**, a gate electrode layer **145**, gate spacer layers **146**, and a gate capping layer **148**.

[0039] The gate dielectric layer **142** may be disposed between the active region ACT and the gate electrode layer **145**. The gate dielectric layer **142** may include oxide, nitride, or a high- κ material. The high- κ material may refer to a dielectric material having a dielectric constant, higher than that of a silicon oxide film (SiO₂).

[0040] The gate electrode layer **145** may be disposed on the gate insulating layer **142**. The gate electrode layer **145** may include a conductive material, for example, a metal nitride such as titanium nitride (TiN), tantalum nitride (TaN), or tungsten nitride (WN), and/or a metal material such as aluminum (Al), tungsten (W), or molybdenum (Mo) or a semiconductor material such as doped polysilicon. The gate electrode layer **145** may include two or more multiple layers.

[0041] The gate spacer layers **146** may be disposed on opposite side surfaces of the gate electrode layer **145**. The gate spacer layers **146** may insulate the source/drain regions **120** and the gate electrode layer **145** from each other. The gate spacer layers **146** may have a multilayer structure in some example embodiments. The gate spacer layers **146** may be formed of oxide, nitride, and oxynitride. In particular, the gate spacer layers **146** may be formed of a low- κ film. For example, the gate spacer layers **146** may include at least one of SiO, SiN, SiCN, SiOC, SiON, and SiOCN.

[0042] The gate capping layer **148** may be disposed on an upper portion of the gate electrode layer **145**, and may have a lower surface and side surfaces, respectively surrounded by the gate electrode layer **145** and the gate spacer layers **146**. The gate capping layer **148** may be formed of, for example, oxide, nitride, and oxynitride.

[0043] Lower interlayer insulating layers **130** may include a first lower interlayer insulating layer **132** covering the source/drain regions **120** and the gate structures **140**, and a second lower interlayer insulating layer **134** disposed on the first lower interlayer insulating layer **132**. The first and second lower interlayer insulating layers **132** and **134** may be distinct layers in a process, and the first and second lower interlayer insulating layers **132** and **134** may be formed of a single layer (not illustrated). The lower interlayer insulating layer **130** may include, for example, at least one of oxide, nitride, and oxynitride, and may include a low- κ material.

[0044] Source/drain contact structures CNT_SD (or “contact structures”) may pass through the first lower interlayer insulating layer **132** to be connected to the source/drain regions **120**, and may apply an electrical signal to the source/drain regions **120**. The source/drain contact structures CNT_SD may be disposed to recess the source/drain regions **120** to a desired (and/or alternatively predetermined) depth, but inventive concepts are not limited thereto. The source/drain contact structures CNT_SD may include a conductive material, for example, a metal material such as tungsten (W), aluminum (Al), copper (Cu), or a semiconductor material such as doped polysilicon. In some example embodiments, the source/drain contact structures CNT_SD may include a barrier metal layer disposed along an external surface thereof. In addition, in some example embodiments, the source/drain contact structures CNT_SD may further include a metal-semiconductor layer, such as a silicide layer, disposed at an interface thereof in contact with the source/drain regions **120**. The source/drain contact structure CNT_SD, disposed on one side of the gate structure **140** and electrically connected to the third line structure **130**, may be referred to as a first source/drain contact structure CNT_SD1. In addition, the source/drain contact structure CNT_SD, disposed on the other side of the gate structure **140**, opposite to the one side of the gate structure **140** and electrically connected to the second line structure **120**, may be referred to as a second source/drain contact structure CNT_SD2.

[0045] The lower vias V0 (or “lower via structures”) may pass through the lower interlayer insulating layers **130** to be connected to the source/drain contact structures CNT_SD. For example, the lower vias V0 may pass through a second lower interlayer insulating layer **164** to be electrically connected to the source/drain contact structure CNT_SD. The lower vias V0 may include a conductive material, for example, a metal material such as tungsten (W), aluminum (Al), or copper (Cu), or a semiconductor material such as doped polysilicon. In some example embodiments, the lower vias V0 may further include a barrier metal layer disposed along an external surface thereof (not illustrated).

[0046] The gate contact structures CNT_G may pass through the lower interlayer insulating layers **130** to be connected to the gate structures **140**. For example, the gate contact structures CNT_G may pass through the first and second lower interlayer insulating layers **132** and **134** and the gate capping layer **148** to be electrically connected to the gate electrode layer **145**. The gate contact structures CNT_G may include a conductive material, for example, a metal material such as tungsten (W), aluminum (Al), copper (Cu), or a semiconductor material such as doped polysilicon. In some example embodiments, the gate contact structures CNT_G may include a barrier metal layer disposed along an external surface thereof.

[0047] Etch stop layers **160** may be disposed on the lower interlayer insulating layers **130**. The etch stop layers **160** may be disposed on, for example, the second lower interlayer insulating layer **134**. The etch stop layers **160** may function as an etch stop layer in an etch process for forming the first interconnection lines M1. The etch stop layers **160** may include at least one of a high- κ material, such as aluminum oxide (Al.sub.2O.sub.3), aluminum nitride (AlN), hafnium oxide (HfO.sub.2), zirconium oxide (ZrO.sub.2), and silicon oxycarbide (SiOC), and combinations thereof. A first etch stop layer **162** and a second etch stop layer **164** on the first etch stop layer **162** may be selected from the group consisting of the above-described high- κ materials, and the first and second etch stop layers **162** and **164** may include different materials. According to an example embodiment, the etch stop layer **160** may be formed of a single layer (see FIGS. 9 and 11).

[0048] The upper interlayer insulating layers **150** may include first and second upper interlayer insulating layers **152** and **154**, sequentially disposed on the etch stop layers **160**. The upper interlayer insulating layer **150** may be formed of silicon oxide or a low- κ material. The upper interlayer insulating layer **150** may include, for example, at least one of SiO, SiN, SiCN, SiOC, SiON, and SiOCN.

[0049] The liner **170** may be disposed between the first and second upper interlayer insulating layers **152** and **154**. The liner **170** may function as an etch stop layer in an etching process for

forming the first interconnection lines **M1**. The liner **170** may include a high- κ material, for example, at least one of aluminum oxide (Al.sub.2O.sub.3), aluminum nitride (AlN), hafnium oxide (HfO.sub.2), and zirconium oxide (ZrO.sub.2). Accordingly, the liner **170** may be referred to as an insulating liner.

[0050] The first interconnection lines **M1** may be interconnections disposed on upper portions of the active regions **ACT** and the gate lines **GL**, and may extend in the X-direction (FIG. 1). The first interconnection lines **M1** may include a first power transmission line **M1_VDD**, a second power transmission line **M1_VSS**, and signal transmission lines **M1_S**. The first power transmission line **M1_VDD** and the second power transmission line **M1_VSS** may be power transmission lines respectively supplying different power voltages (**VDD** and **VSS**) to a semiconductor device, and may be electrically connected to source/drain regions on the active regions **ACT**. The signal transmission lines **M1_S** may be signal transmission lines supplying a signal to the semiconductor device, and may be electrically connected to the gate electrode **GL_G**.

[0051] First interconnection lines **180** may be interconnections disposed on the upper portions of the active regions **ACT** and the gate lines **GL** (FIG. 2), and the first interconnection lines **180** may be disposed to correspond to the first interconnection lines **M1** of FIG. 1.

[0052] The first interconnection lines **180** may include first to third line structures **181**, **182**, and **183** disposed to extend in the first direction (X-direction) and to be spaced apart from each other in the second direction (Y-direction). The first and second line structures **181** and **182** may be components corresponding to the signal transmission lines **M1_S**. Accordingly, the first line structure **181** may be referred to as a first signal transmission line structure, and the second line structure **182** may be referred to as a second signal transmission line structure. The third line structure **183** may be a component corresponding to the first power transmission line **M1_VDD** or the second power transmission line **M1_VSS**.

[0053] Referring to FIGS. 1, 3, and 5, each of the first to third line structures **181**, **182**, and **183** may include a line portion (or “interconnection portion”) extending in the first direction (X-direction), and an extension portion (or “protrusion portion”) extending vertically downwardly from at least a portion of a lower surface of the line portion.

[0054] The first line structure **181** may include a first line portion **181a** extending in the first direction (X-direction) across the gate structure **140** at the top of the gate structure **140**, and a first extension portion **181b** vertically extending from at least a portion of a lower surface of the first line portion **181a** toward the gate structure **140**, on the gate structure **140**.

[0055] The first line portion **181a** may be defined as a component passing through at least a portion of the second upper interlayer insulating layer **154** and the insulating liner **170**, and extending in the first direction (X-direction) from an upper surface of the first upper interlayer insulating layer **152**.

[0056] The first extension portion **181b** may be defined as a component vertically downwardly protruding from at least a portion of the lower surface of the first line portion **181a** toward the gate structure **140**. Accordingly, the first extension portion **181b** may be referred to as a first protrusion portion. The first extension portion **181b** may pass through the first upper interlayer insulating layer **152** and the etch stop layers **162** and **164**, and may have a lower surface in contact with an upper surface of the gate contact structure **CNT_G**. A horizontal width of the lower surface of the first extension portion **181b** in the second direction (Y-direction) may be substantially equal to a horizontal width of the upper surface of the gate contact structure **CNT_G** in the second direction (Y-direction), but inventive concepts are not limited thereto.

[0057] The first line portion **181a** and the first extension portion **181b** may include the same conductive material. Accordingly, the lower surface of the first line portion **181a** and an upper surface of the first extension portion **181b** may not be distinguished from each other in terms of a boundary therebetween. The first line portion **181a** and the first extension portion **181b** may include, for example, at least one of aluminum (Al), copper (Cu), and tungsten (W). In inventive

concepts, the first line portion **181a** and the first extension portion **181b** may include copper (Cu). [0058] A first vertical thickness **d1** of the first line portion **181a** may be greater than a second vertical thickness **d2** of the first extension portion **181b**. A ratio of the first vertical thickness **d1** and the second vertical thickness **d2** may be 2:1 or more, for example, 2:1 to 6:1, or 3:1 to 5:1. [0059] The third line structure **183** may include a third line portion **183a** extending in the first direction (X-direction) across the gate structure **140** and the source/drain contact structure CNT_SD at the top of the gate structure **140**, and a third extension portion **183b** vertically extending from at least a portion of a lower surface of the third line portion **183a** toward a via structure **V0**, on the source/drain contact structure CNT_SD.

[0060] The third line portion **183a** may be defined as a component passing through at least a portion of the second upper interlayer insulating layer **154** and the insulating liner **170**, and extending in the first direction (X-direction) from the upper surface of the first upper interlayer insulating layer **152**.

[0061] The third extension portion **183b** may be defined as a component extending vertically downwardly from at least a portion of the lower surface of the third line portion **183a** toward the via structure **V0**. The third extension portion **183b** may pass through the first upper interlayer insulating layer **152** and the etch stop layers **162** and **164**, and may have a lower surface in contact with an upper surface of the via structure **V0**. A horizontal width of a lower surface of the third extension portion **183b** in the second direction (Y-direction) may be greater than a horizontal width of the upper surface of the via structure **V0** in the second direction (Y-direction).

[0062] The third line portion **183a** and the third extension portion **183b** may include the same conductive material. Accordingly, a lower surface of the first line portion **183a** and an upper surface of the third extension portion **183b** may not be distinguished from each other in terms of a boundary therebetween. For example, the third line portion **183a** and the third extension portion **183b** may include at least one of aluminum (Al), copper (Cu), and tungsten (W). In inventive concepts, the third line portion **183a** and the third extension portion **183b** may include copper (Cu). [0063] A vertical thickness of the third line portion **183a** may be substantially equal to the first vertical thickness **d1** of the first line portion **181a**. Similarly, a vertical thickness of the third extension portion **183b** may be substantially equal to the second vertical thickness **d2** of the first extension portion **181b**. Accordingly, a ratio of the vertical thickness of the third line portion **183a** and the vertical thickness of the third extension portion **183b** may correspond to the first vertical thickness **d1** of the first line portion **181a** and the second vertical thickness **d2** of the first extension portion **181b**.

[0064] The second line structure **182** may be disposed between the first line structure **181** and the third line structure **183**, and may extend in the first direction (X-direction) across the gate structure **140** and the source/drain contact structure CNT_SD at the top of the gate structure **140**.

[0065] In a similar manner to the first line structure **181** and the third line structure **183**, the second line structure **182** may also include a second line portion **182a** extending in the first direction (X-direction) across the gate structure **140** and the source/drain contact structure CNT_SD at the top of the gate structure **140**, and a second extension portion **182b** vertically extending from at least a portion of a lower surface of the second line portion **182a** toward the via structure **V0**, on the source/drain contact structure CNT_SD (see FIGS. 1 and 3).

[0066] Here, the second extension portion **182b** of the second line structure **182** and the third extension portion **183b** of the third line structure **183** may be positioned on opposite sides of the gate structure **140**. For example, the second extension portion **182b** of the second line structure **182** may be positioned on one side of the gate structure **140**, and the third extension portion **183b** of the third line structure **183** may be positioned on the other side of the gate structure **140**, opposite to the one side of the gate structure **140**.

[0067] The second line portion **182a** may be defined as a component passing through at least a portion of the second upper interlayer insulating layer **154** and the insulating liner **170**, and

extending in the first direction (X-direction) from the upper surface of the first upper interlayer insulating layer **152**.

[0068] The second extension portion **182b** may be defined as a component vertically downwardly protruding from at least a portion of the lower surface of the second line portion **182a** toward the via structure **V0**. The second extension portion **182b** may pass through the first upper interlayer insulating layer **152** and the etch stop layers **162** and **164**, and may have a lower surface in contact with an upper surface of the source/drain contact structure CNT_SD. A horizontal width of the lower surface of the second extension portion **182b** in the second direction (Y-direction) may be substantially equal to a horizontal width of the upper surface of the via structure **V0** in the second direction (Y-direction).

[0069] The second line portion **182a** and the second extension portion **182b** may include the same conductive material. Accordingly, the lower surface of the second line portion **182a** and an upper surface of the second extension portion **182b** may not be distinguished from each other in terms of a boundary therebetween. The second line portion **182a** and the second extension portion **182b** may include, for example, at least one of aluminum (Al), copper (Cu), and tungsten (W). In inventive concepts, the second line portion **182a** and the second extension portion **182b** may include copper (Cu).

[0070] In at least one cross-sectional view, among a plurality of cross-sectional views extending in the second direction (Y-direction), the plurality of cross-sectional views in the third direction (Z-direction), levels of lowermost surfaces of the first to third line structures **181**, **182**, and **183** may be different from each other.

[0071] For example, in a first cross-sectional view extending in the second direction, the first cross-sectional view in a third direction overlapping the upper surface of the gate contact structure CNT_G, among the third directions, the lowermost surface of the first line structure **181** may be on a level, lower than those of the lowermost surface of the second line structure **182** and/or the lowermost surface of the third line structure **183** (see the cross-section taken along I-I' of FIG. 3). Such a configuration may be due to the first extension portion **181b** of the first line structure **181**. Accordingly, a level difference between the lowermost surface of the first line structure **181**, and the lowermost surfaces of the second line structure **182** and/or the third line structure **183** may be the vertical thickness d2 of the first extension portion **181b**.

[0072] According to inventive concepts, a sufficient separation distance s1 may be secured between the first line structure **181** and the second line structure **182**, adjacent to each other. Here, the separation distance s1 may be defined as a minimum distance between an outer periphery of the lowermost surface of the first line structure **181** and an outer periphery of the lowermost surface of the second line structure **182**. The sufficient secured separation distance s1 may be achieved by the first extension portion **181b** of the first line structure **181**.

[0073] Accordingly, a short circuit that may occur between the second line structure **182** and the gate contact structure CNT_G, positioned on a lower portion of the first line structure **181**, may be limited and/or prevented and/or minimized.

[0074] In addition, in a second cross-sectional view extending in the second direction, the second cross-sectional view in a third direction overlapping an upper surface of the first source/drain contact structure CNT_SD1, among the third directions, the lowermost surface of the third line structure **183** may be on a level, lower than those of the lowermost surfaces of the first line structure **181** and/or the second line structure **182** (see the cross-sections taken along II-II' of FIGS. 1 and 3). Such a configuration may be due to the third extension portion **183b** of the third line structure **183**. Accordingly, a level difference between the lowermost surface of the third line structure **183** and the lowermost surfaces of the first line structure **181** and/or the second line structure **182** may be the vertical thickness d2 of the third extension portion **183b**.

[0075] Accordingly, a sufficient separation distance s2 may be secured between the second line structures **182** and the third line structures **183**, adjacent to each other. Here, the separation distance

s2 may be defined as a minimum distance between an outer peripheral surface of the lowermost surface of the second line structure **182** and an outer peripheral surface of the lowermost surface of the third line structure **183**. The sufficient secured separation distance **s2** may be achieved by the third extension portion **183b** of the third line structure **183**.

[0076] Accordingly, a short circuit that may occur between the second line structure **182** and the source/drain contact structure CNT_SD, positioned on a lower portion of the third line structure **183**, may be limited and/or prevented and/or minimized.

[0077] In addition, a vertical depth of the third extension portion **183b** of the third line structure **183** in the third direction (Z-direction) may be secured without risk to the short circuit.

[0078] Although not illustrated, in a third cross-sectional view extending in the second direction, the third cross-sectional view in a third direction overlapping an upper surface of the second source/drain contact structure CNT_SD2, among the third directions, the lowermost surface of the second line structure **182** may be on a level, lower than those of the lowermost surfaces of the first line structure **181** and/or the third line structure **183**. Such a configuration may be due to the second extension portion **182b** of the second line structure **182** (see FIG. 1).

[0079] Referring to FIGS. 1, 4, and 5, the first extension portion **181b** of the first line structure **181**, the second extension portion **182b** of the second line structure **182**, and the third extension portion **183b** of the third line structure **183** may have a cylindrical shape. In other words, outer peripheries of the first extension portion **181b**, the second extension portion **182b**, and the third extension portion **183b** may have a circular cross-section.

[0080] A horizontal width of the outer periphery of the first extension portion **181b** may be substantially equal to a first horizontal width **w1** of the first line portion **181a** in the second direction (Y-direction). Similarly, a horizontal width of the outer periphery of the second extension portion **182b** may be substantially equal to a second horizontal width **w2** of the second line portion **182a** in the second direction (Y-direction). Here, the first and second horizontal widths **w1** and **w2** may be substantially equal to each other, but inventive concepts are not limited thereto.

[0081] Similarly, a horizontal width of the outer periphery of the third extension portion **183b** may be substantially equal to a third horizontal width **w3** of the third line portion **183a** in the second direction (Y-direction). Here, the third horizontal width **w3** may be greater than the first and second horizontal widths **w1** and **w2**.

[0082] Referring back to FIGS. 2 and 3, the insulating liner **170** may extend along a side surface of an extension portion of each of the first to third line structures **181**, **182**, and **183**.

[0083] For example, the insulating liner **170** may be in contact with a side surface of a lower region of the first line portion **181a**, and may vertically downwardly extend along a side surface of the first extension portion **181b** and a side surface of a first upper insulating layer **152** (see the cross-section taken along I-I' in FIG. 2 and FIG. 3).

[0084] In addition, the insulating liner **170** may be in contact with a side surface of a lower region of the third line portion **183a**, and may vertically downwardly extend along a side surface of the third extension portion **183b** and the side surface of the first upper insulating layer **152** (see the cross-section taken along II-II' in FIG. 2 and FIG. 3).

[0085] Although not illustrated, at least a portion of the insulating liner **170** may be in contact with a side surface of a lower region of the second line portion **182a**, and may vertically downwardly extend along a side surface of the second extension portion **182b** and the side surface of the first upper insulating layer **152**.

[0086] FIGS. 7 to 11 are partially enlarged views of a semiconductor device according to example embodiments.

[0087] Referring to FIG. 7, a semiconductor device **100a** may be the same as or similar to that illustrated in FIGS. 1 to 6, except that upper regions of first to third line structures **181**, **182**, and **183** have a tapering structure.

[0088] Referring to FIG. 7, at least a portion of a side surface **181a_SS** of a first line portion **181a**

may include a portion inclined with respect to a lower surface of the first line portion **181a**. In other words, a horizontal width **w1** of the first line portion **181a** in a second direction (Y-direction) may gradually decrease as a distance to a first extension portion **181b** decreases.

[0089] Similarly, a horizontal width **w3** of a third line portion **183a** of the third line structure **183** in the second direction (Y-direction) may also gradually decrease as a distance to a third extension portion **183b** decreases. Although not illustrated, the above-described features may be applied to the second line structure in the same manner.

[0090] Referring to FIG. **8**, a semiconductor device **100b** may be the same as or similar to that illustrated in FIGS. **1** to **6**, except that upper regions of first to third line structures **181**, **182**, and **183** have a step.

[0091] Referring to FIG. **8**, a lower surface **181a_LS** of a first line portion **181a** may have a step. For example, at least a portion of the lower surface **181a_LS** of the first line portion **181a** may be in contact with at least a portion of an upper surface of an insulating liner **170**.

[0092] Similarly, a lower surface **183a_LS** of a third line portion **183a** may have a step. For example, at least a portion of the lower surface **183a_LS** of the third line portion **183a** may be in contact with at least a portion of the upper surface of the insulating liner **170**. Although not illustrated, the above-described features may be applied to the second line structure in the same manner.

[0093] Referring to FIG. **9**, a semiconductor device **100c** may be the same as or similar to those illustrated in FIGS. **1** to **8**, except that an insulating liner **170** is in contact with only side surfaces of first to third line structures **181**, **182**, and **183**.

[0094] Referring to FIG. **9**, the insulating liner **170** may be in contact with only side surfaces of lower regions of first and third line portions **181a** and **183a** of the first and third line structures **181** and **183**, on a first upper interlayer insulating layer **152**. In another view, the insulating liner **170** may not extend to a space between first and third extension portions **181b** and **183b** of the first and third line structures **181** and **183** and the first upper interlayer insulating layer **152**. Although not illustrated, the above-described features may be applied to the second line structure in the same manner.

[0095] Referring to FIG. **10**, a semiconductor device **100d** may be the same as or similar to those illustrated in FIGS. **1** to **8**, except that the semiconductor device **100d** further includes a barrier layer BL disposed along external surfaces of conductive material layers of the first to third line structures **181**, **182**, and **183**.

[0096] Referring to FIG. **10**, the barrier layer BL may be additionally disposed on at least a portion of external surfaces of the first and third line structures **181** and **183**. For example, the barrier layer BL may be disposed on side surfaces of first and third line portions **181a** and **183a**, and may extend to cover side surfaces and lower surfaces of the first and third extension portions **181b** and **183b**. The barrier layer BL may include at least one of titanium (Ti), tantalum (Ta), cobalt (Co), titanium nitride (TiN), and tantalum nitride (TaN). Although not illustrated, the above-described features may be applied to the second line structure in the same manner.

[0097] Referring to FIG. **11**, a semiconductor device **100e** may be the same or similar to those described with reference to FIGS. **1** to **9**, except that an etch stop layer **160** is a single layer.

[0098] Referring to FIG. **11**, the etch stop layer **160** between a second lower interlayer insulating layer **134** and a first upper interlayer insulating layer **152** may be a single layer. The etch stop layer **160** may include a material the same as or similar to that of the etch stop layer **160** described with reference to FIGS. **1** to **6**.

[0099] FIG. **12** is a cross-sectional view of a semiconductor device according to example embodiments. FIG. **12** illustrates a schematic cross-section of the semiconductor device of FIG. **1**, taken along line V-V'.

[0100] Referring to FIG. **12**, a semiconductor device **100f** may further include a plurality of channel layers **115** disposed on an active regions ACT to be vertically spaced from each other, and

internal spacer layers **118** disposed between the plurality of channel layers **115** to be parallel to a gate electrode layer **145**. The semiconductor device **100f** may include transistors having a gate-all-around structure in which a gate structure **140a** is disposed between an active fin and the channel layers **115** and between the plurality of channel layers **115** having a nano sheet shape. For example, the semiconductor device **100f** may include transistors having a multi-bridge-channel FET (MBCFETTM) structure, formed by the channel layers **115**, source/drain regions **120**, and the gate structure **140a**.

[0101] The plurality of channel layers **115** may be disposed on the active fin as two or more layers disposed to be spaced apart from each other in a direction, perpendicular to an upper surface of the active fin, for example, a Z-direction. The channel layers **115** may be connected to the source/drain regions **120**, and may be spaced apart from an upper surface of an active fin **105**. The channel layers **115** may have a width equal to or similar to that of the active fin in a Y-direction, and may have a width equal to or similar to that of the gate structure **140a** in an X-direction. However, in some example embodiments, the channel layers **115** may have a reduced width such that side surfaces of the channel layers **115** are positioned on a lower portion of the gate structure **140a** in the X-direction.

[0102] The plurality of channel layers **115** may be formed of a semiconductor material, and may include, for example, at least one of silicon (Si), silicon germanium (SiGe), and germanium (Ge). For example, the channel layers **115** may be formed of a material the same as that of as the substrate **101**. In example embodiments, the number and shape of the channel layers **115**, included in a single channel structure, may be changed in various manners. For example, in some example embodiments, a channel layer may be further positioned in a region in which active fins **105** are in contact with the gate electrode layer **145**.

[0103] The gate structure **140a** may be disposed on upper portions of the active fins **105** and the plurality of channel layers **115**, and may extend to intersect the active fins **105** and the plurality of channel layers **115**. Channel regions of transistors may be formed in the active fins **105** and the plurality of channel layers **115**, intersecting the gate structure **140a**. In present example embodiment, a gate insulating layer **142** may be disposed not only between the active fin **105** and the gate electrode layer **145**, but also between the plurality of channel layers **115** and the gate electrode layer **145**. The gate electrode layer **145** may be disposed on upper portions of the active fins **105** to extend to the upper portions of the plurality of channel layers **115** while filling a space between the plurality of channel layers **115**. The gate electrode layer **145** may be spaced apart from the plurality of channel layers **115** by the gate insulating layer **142**.

[0104] The internal spacer layers **118** may be disposed between the plurality of channel layers **115** to be parallel to the gate electrode layer **145**. The gate electrode layer **145** may be spaced apart from the source/drain regions **120** by the internal spacer layers **118** to be electrically isolated from the source/drain regions **120**. The internal spacer layers **118** may have a flat side surface opposing the gate electrode layer **145**, or may have an inwardly convex rounded shape toward the gate electrode layer **145**. The internal spacer layers **118** may be formed of oxide, nitride, and oxynitride. In particular, the internal spacer layers **118** may be formed of a low- κ film.

[0105] In example embodiments, the semiconductor device **100f** having the MBCFETTM structure may be disposed in one region of the semiconductor device described above with reference to FIGS. 7 to 11. In addition, in example embodiments, at least one region of the semiconductor device may include a vertical FET including an active region extending to be perpendicular to an upper surface of a substrate **101** and a gate structure disposed to surround the active region.

[0106] FIGS. 13 to 22 are vertical cross-sectional views of sequential processes of a method of manufacturing a semiconductor device according to an example embodiment.

[0107] FIGS. 13 to 15 may be process diagrams of sequential processes of forming a middle of line (MOL) of the semiconductor device **100** according to an example embodiment.

[0108] Referring to FIG. 13, a substrate **101**, a device isolation layer **110** defining an active region

ACT, on the substrate **101**, a gate structure **140** on the active region ACT, a source/drain region **120** formed on at least one side of the gate structure **140**, on the active region ACT, and a first lower interlayer insulating layer **132** covering the gate structure **140** and the source/drain region **120**.

[0109] The device isolation layer **110** may be formed by patterning the substrate **101** to form a desired (and/or alternatively predetermined) trench defining an active fin, and then filling the trench with an insulating material. Thereafter, a portion of the insulating material may be removed to allow an upper region of the active fin to protrude toward the substrate **101**.

[0110] Thereafter, a dummy gate insulating layer and a dummy gate electrode, extending to intersect the active fin, may be sequentially stacked, and then a spacer **146** may be formed on opposite sides of the dummy gate insulating layer and the dummy gate electrode (see the spacer **146** in FIG. **6**). Subsequently, a portion of the active fin may be selectively removed from opposite sides of the spacer **146**.

[0111] Subsequently, referring to FIG. **6**, the source/drain region **120** may be formed on the recessed active fin on the opposite sides of the spacer **146**. The source/drain region **120** may be formed using, for example, a selective epitaxial growth (SEG) process. The source/drain region **120** may be grown along a crystallographically stable surface during the growth process, and thus may have a pentagonal or hexagonal shape, as illustrated. However, a size and shape of the source/drain region **120** are not limited to those illustrated in the drawings.

[0112] Thereafter, the dummy gate insulating layer and the dummy gate electrode may be removed to sequentially form a gate insulating layer **142**, a gate electrode layer **145**, and a gate capping layer **148**, and then the first lower interlayer insulating layer **132**, covering the gate structure **140** and the source/drain region **120**, may be formed.

[0113] After the first lower interlayer insulating layer **132** is first formed, a planarization process may be performed on an upper surface of the dummy gate electrode, and then the dummy gate insulating layer and the dummy gate electrode may be removed to form the gate structure **140**. In this case, the first lower interlayer insulating layer **132** may not be disposed on the gate structure **140**.

[0114] Referring to FIG. **14**, the first lower interlayer insulating layer **132** may be patterned to form source/drain contact structures CNT_SD in contact with the source/drain region **120**.

[0115] A desired (and/or alternatively predetermined) mask layer may be formed on the first lower interlayer insulating layer **132**, and then openings, exposing an upper region of the source/drain region **120**, may be formed using the mask layer. The desired (and/or alternatively predetermined) mask layer may be, for example, a photoresist layer.

[0116] The source/drain contact structures CNT_SD may be formed by filling a conductive material in the openings.

[0117] Referring to FIG. **15**, a gate contact structure CNT_G, physically connected to the gate structure **140**, and a via structure V0, physically connected to the source/drain contact structures CNT_SD, may be formed.

[0118] A second lower interlayer insulating layer **134** may be formed on the first lower interlayer insulating layer **132** and the source/drain contact structures CNT_SD. Thereafter, a desired (and/or alternatively predetermined) mask layer may be formed on the second lower interlayer insulating layer **134**, and then openings, exposing at least a portion of the gate electrode layer **145**, and openings, exposing at least a portion of the source/drain contact structure CNT_SD, may be formed using the mask layer.

[0119] The gate contact structure CNT_G and the via structure V0 may be formed by filling a conductive material in the openings.

[0120] FIGS. **16** to **22** may be process diagrams of sequential processes of forming a back end of line (BEOL) of the semiconductor device **100** according to an example embodiment. FIG. **16** may be a process diagram following FIG. **15**.

[0121] Referring to FIG. **16**, etch stop layers **160** and a first upper interlayer insulating layer **152**

may be sequentially formed on the second lower interlayer insulating layer **134**.

[0122] A first etch stop layer **162** and a second etch stop layer **164**, covering the gate contact structure CNT_G and the source/drain contact structures CNT_SD, may be sequentially formed on the second lower interlayer insulating layer **134**. Each of the first and second etch stop layers **162** and **164** may be selected from the group consisting of aluminum oxide (Al.sub.2O.sub.3), aluminum nitride (AlN), hafnium oxide (HfO.sub.2), zirconium oxide (ZrO.sub.2), and silicon oxycarbide (SIOC). In the present example embodiment, the first etch stop layer **162** may include aluminum oxide (Al.sub.2O.sub.3), and the second etch stop layer **162** may include silicon oxycarbide (SIOC).

[0123] The first upper interlayer insulating layer **152**, covering the etch stop layers **160**, may be formed. The first upper interlayer insulating layer **152** may include, for example, at least one of oxide, nitride, and oxynitride.

[0124] Referring to FIG. **17**, a first mask layer M1 having a plurality of first open regions P1 may be formed on the first upper interlayer insulating layer **152**.

[0125] The first mask layer M1 may be aligned such that the plurality of first open regions P1 are positioned on the gate contact structure CNT_G and the via structure V0.

[0126] The first mask layer M1 may be, for example, a photoresist layer. At least a portion of an upper surface of the first upper interlayer insulating layer **152** may be exposed by the first open region P1.

[0127] Referring to FIG. **18**, the first upper interlayer insulating layer **152** may be patterned using the first mask layer M1.

[0128] The first upper interlayer insulating layer **152**, exposed by the first open region P1, may be removed, and the etch stop layers **160**, exposed by removing the first upper interlayer insulating layer **152**, may be removed to form a first etched region OP1.

[0129] Referring to FIG. **1** together with FIG. **18**, the first etched region OP1 may be in the form of a hole. In addition, a horizontal width of the first etched region OP1, formed on the gate contact structure CNT_G, may be greater than a horizontal width of the first etched region OP1, formed on the via structure V0.

[0130] Referring to FIG. **19**, a liner **170** may be formed on the first upper interlayer insulating layer **152**.

[0131] The liner **170** may be conformally formed along the upper surface of the first upper interlayer insulating layer **152** and an external surface of the first etched region OP1. The liner **170** may include at least one of aluminum oxide (Al.sub.2O.sub.3), aluminum nitride (AlN), hafnium oxide (HfO.sub.2), and zirconium oxide (ZrO.sub.2).

[0132] Referring to FIG. **20**, a second upper interlayer insulating layer **154** may be formed on the liner **170**.

[0133] The second upper interlayer insulating layer **154** may be formed by forming an insulating material on the liner **170** and performing a planarization process. The second upper interlayer insulating layer **154** may include a material the same as that of the first upper interlayer insulating layer **152**.

[0134] Referring to FIG. **21**, a second mask layer M2, having a plurality of second open regions P2, may be formed on the second upper interlayer insulating layer **154**.

[0135] The second mask layer M2 may be aligned such that at least a portion of the second open regions P2 is positioned on the first etched region OP1. In this case, at least a portion of the second open regions P2 may have a horizontal width substantially equal to a horizontal width of the first etched region OP1. Here, the horizontal width may refer to a horizontal width in a second direction, for example, a Y-direction (see FIGS. **1** and **2**).

[0136] At least a portion of an upper surface of the second upper interlayer insulating layer **154** may be exposed by the second open region P2.

[0137] Referring to FIG. **22**, the second upper interlayer insulating layer **154** may be patterned

using the second mask layer M2.

[0138] The second upper interlayer insulating layer **154**, exposed by the second open region P2, may be removed, and the insulating liner **170**, exposed by removing the second upper interlayer insulating layer **152**, may be removed to form a second etched region OP2.

[0139] Referring to FIG. 1 together with FIG. 22, the second etched region OP2 may be in the form of a line. In addition, a horizontal width of the second etched region OP2, formed on the gate contact structure CNT_G, may be greater than a horizontal width of the second etched region OP2, formed on the via structure V0. Here, the horizontal width may refer to a horizontal width in the second direction, for example, the Y-direction (see FIGS. 1 and 2).

[0140] Thereafter, the second etched region OP2 may be filled with a conductive material to form first interconnection lines **180**, specifically, first to third line structures **181**, **182**, and **183** (see FIGS. 1 and 2).

[0141] Subsequently, although not illustrated, an interlayer insulating layer covering the first interconnection lines **180**, on the second upper interlayer insulating layer **154**, may be additionally formed, and an upper via passing through the interlayer insulating layer, the upper via connected to the first interconnection lines **180**, and second interconnection lines, electrically connecting the first interconnection lines **180** to each other, may be formed on the interlayer insulating layer. If necessary, a process of forming an interlayer insulating layer, upper via, and interconnection lines may be repeated in the same manner.

[0142] According to example embodiments of inventive concepts, a semiconductor device may include interconnection lines respectively having a step in one direction to limit and/or prevent a short circuit between the interconnection lines.

[0143] Specifically, the semiconductor device according to inventive concepts may include interconnection lines having a step in a first direction (for example, an X-direction). Due to the step, levels of lowermost surfaces of the interconnection lines may be different from each other in at least one cross-sectional view among a plurality of cross-sectional views extending in a second direction (for example, a Y-direction), the plurality of cross-sectional views in a third direction (for example, a Z-direction). Accordingly, a sufficient separation distance may be secured between adjacent interconnection lines, thereby limiting and/or minimizing and/or preventing a short circuit.

[0144] While example embodiments have been shown and described above, it will be apparent to those skilled in the art that modifications and variations could be made without departing from the scope of inventive concepts as defined by the appended claims.

Claims

1. A semiconductor device comprising: a substrate including an active region extending in a first direction; a gate structure on the active region and intersecting the active region, the gate structure including a gate electrode and a gate capping layer on the gate electrode, the gate structure extending in a second direction, the second direction intersecting the first direction; a source/drain region on the active region, the source/drain region on a side of the gate structure; a contact structure on the source/drain region and electrically connected to the source/drain region; a first insulating layer on the gate structure and the contact structure; a gate contact structure passing through the first insulating layer and the gate capping layer of the gate structure, the gate contact structure electrically connected to the gate electrode of the gate structure; a via structure passing through the first insulating layer, the via structure being electrically connected to the contact structure; an etch stop layer on the first insulating layer and the gate contact structure; a second insulating layer on the etch stop layer; an insulating liner on the second insulating layer; and interconnection lines on the first insulating layer, the interconnection lines including a first line, a second line, and a third line sequentially spaced apart in the second direction on the first insulating layer, wherein each of the first line, the second line, and the third line include a line portion and an

extension portion extending from the line portion in a third direction, in each of the first line, the second line, and the third line, the line portion passes through the insulating liner and extends in the first direction on the second insulating layer, the third direction is perpendicular to the first direction and the second direction, and the extension portion of the first line, the extension portion of the second line, and the extension portion of the third line each pass through the second insulating layer and the etch stop layer and are electrically connected to a corresponding one of the gate contact structure and the via structure.

2. The semiconductor device of claim 1, wherein the first line is a signal transmission line electrically connected to the gate electrode of the gate structure, and the third line is a power transmission line electrically connected to the source/drain region.

3. The semiconductor device of claim 1, wherein in each of the first line, the second line, and the third line, the line portion and the extension portion connected to the line portion include a same conductive material.

4. The semiconductor device of claim 1, wherein in each of the first line, the second line, and the third line, a thickness of the extension portion is $\frac{1}{5}$ to $\frac{1}{3}$ of a thickness of the line portion connected to the extension portion.

5. The semiconductor device of claim 1, wherein in each of the first line, the second line, and the third line, the extension portion has a cylindrical shape.

6. The semiconductor device of claim 1, wherein a lower surface of the extension portion of the first line is in contact with an upper surface of the gate contact structure, and in a cross-section in the third direction, which overlaps the upper surface of the gate contact structure, a level of the lower surface of the extension portion of the first line is lower than a level of a lower surface of the second line and a level of a lower surface of the third line.

7. The semiconductor device of claim 1, wherein a lower surface of the extension portion of the third line is in contact with an upper surface of the contact structure, and in a cross-section in the third direction, which overlaps the upper surface of the contact structure, a level of the lower surface of the third line is lower than a level of the lower surface of the first line and a level of the lower surface of the second line.

8. The semiconductor device of claim 1, wherein a lower surface of the extension portion of the first line is in contact with an upper surface of the gate contact structure, a lower surface of the extension portion of the third line is in contact with an upper surface of the contact structure, in a first cross-section extending in the second direction, which overlaps the upper surface of the gate contact structure, a level of the lower surface of the first line is lower than a level of a lower surface of the second line and the lower surface of the third line, and in a second cross-section extending in the second direction, which overlaps the upper surface of the contact structure, the lower surface of the third line is lower than the lower surface of the first line and the lower surface of the second line.

9. The semiconductor device of claim 8, wherein the insulating liner is in contact with a side surface of the line portion of the first line on the second insulating layer and a side surface of the line portion of the third line on the second insulating layer, in the first cross-section, the insulating liner extends from the side surface of the line portion of the first line to a space between the extension portion of the first line and the second insulating layer, and in the second cross-section, the insulating liner extends from the side surface of the line portion of the third line to a space between the extension portion of the third line and the second insulating layer.

10. The semiconductor device of claim 8, wherein in each of the first line, the second line, and the third line, a horizontal width of the line portion in the second direction gradually decreases as a distance to an upper surface of the extension portion, corresponding to the line portion, decreases.

11. The semiconductor device of claim 8, wherein in each of the first line, the second line, and the third line, a horizontal width in the second direction of an upper region of the line portion is greater than a horizontal width in the second direction of the extension portion corresponding to the line

portion, and in each of the first line, the second line, and the third line, a lower surface of a portion of the upper region of the line portion is in contact with an upper surface of the insulating liner.

12. The semiconductor device of claim 8, wherein an external surface of the extension portion of the first line, an external surface of the extension portion of the second line, and an external surface of the extension portion of the third line are in contact with the second insulating layer and a corresponding side surface of the etch stop layer.

13. The semiconductor device of claim 8, further comprising: a barrier layer covering a side surface of the line portion of the first line, a side surface of the line portion of the second line, and a side surface of the line portion of the third line, wherein the barrier layer extends along a side surface of the extension portion of the first line, a lower surface of the extension portion of the first line, a side surface of the extension portion of the second line, a lower surface of the extension portion of the second line, a side surface of the extension portion of the third line, and a lower surface of the extension portion of the third line.

14. The semiconductor device of claim 1, wherein the etch stop layer includes a plurality of etch stop layers, and the plurality of etch stop layers include a first layer and a second layer on the first layer, and a material of the second layer is different from a material of the first layer.

15. The semiconductor device of claim 1, further comprising: a plurality of channel layers on the active region and spaced apart from each other in a vertical direction, wherein the vertical direction is perpendicular to an upper surface of the substrate, and the gate structure surrounds the plurality of channel layers.

16. A semiconductor device comprising: a substrate including an active region extending in a first direction; a gate structure on the active region and intersecting the active region, the gate structure extending in a second direction, the second direction intersecting the first direction; a source/drain region on the active region, the source/drain region on a side of the gate structure; a contact structure on the source/drain region and electrically connected to the source/drain region; a via structure on the contact structure and electrically connected to the contact structure; a gate contact structure on the gate structure and electrically connected to the gate structure; a first signal transmission line structure including a first line portion extending in one direction across the gate structure and a first protrusion portion vertically protruding from a lower surface of the first line portion toward the gate contact structure, the first signal transmission line structure being on the gate structure; and a power transmission line structure including an interconnection portion and an extension portion, the interconnection portion extending in the one direction across the gate structure and the contact structure, and the extension portion vertically extending from a lower surface of the interconnection portion toward the via structure, the power transmission line structure being on the contact structure.

17. The semiconductor device of claim 16, further comprising: a second signal transmission line structure between the first signal transmission line structure and the power transmission line structure, the second signal transmission line structure including a second line portion extending in the one direction across the gate structure and the contact structure, the second signal transmission line being on the active region, wherein a level of a lower surface of the second line portion of the second signal transmission line structure is higher than a level of a lower surface of the first protrusion portion of the first signal transmission line structure and a level of a lower surface of the extension portion of the power transmission line structure.

18. The semiconductor device of claim 17, wherein the second signal transmission line structure further includes a second protrusion portion vertically protruding from the lower surface of the second line portion toward the via structure, on the contact structure, an outer periphery of each of the first protrusion portion, the second protrusion portion, and the extension portion has a circular cross-section.

19. A semiconductor device comprising: a substrate including an active region extending in a first direction; a gate structure on the active region and intersecting the active region, the gate structure

including a gate electrode and a gate capping layer on the gate electrode, the gate structure extending in a second direction, the second direction intersecting the first direction; a source/drain region on the active region, the source/drain region on a side of the gate structure; a contact structure on the source/drain region and electrically connected to the source/drain region; a first insulating layer on the gate structure and the contact structure; a gate contact structure passing through the first insulating layer and the gate capping layer of the gate structure, the gate contact structure electrically connected to the gate electrode of the gate structure; a via structure passing through the first insulating layer, the via structure electrically connected to the contact structure; an etch stop layer on the first insulating layer and the gate contact structure; a second insulating layer on the etch stop layer; an insulating liner on the second insulating layer; and first interconnection lines on the first insulating layer and spaced apart from each other in the second direction, wherein the first interconnection lines include a first line structure and a second line structure, the first line structure includes a first line portion and a protrusion portion, the first line portion is on the second insulating layer and extends in one direction across the gate structure, the protrusion portion vertically protrudes from a lower surface of the first line portion, passes through the second insulating layer and the etch stop layer, and is in contact with the gate contact structure, the second line structure includes an interconnection portion and an extension portion, the interconnection portion is on the second insulating layer and extends in the one direction across the gate structure and the contact structure, the extension portion vertically protrudes from a lower surface of the interconnection portion, passes through the second insulating layer and the etch stop layer, and is in contact with the via structure, and outer peripheries of the protrusion portion of the first line structure and the extension portion of the second line structure have a circular cross-section.

20. The semiconductor device of claim 8, further comprising: a barrier layer extending from a side surface of the first line portion to a side surface of the protrusion portion.
